

A Fixed-Kernel Obstruction for Rank-Two Norm-Attaining Operators on Martin's Space

FA/Banach discovery run `fa_banach_001`

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Source and Open Problem

This packet concerns the space constructed in:

Miguel Martin, *A Banach space whose set of norm-attaining functionals is algebraically trivial*, arXiv:2406.07273.

The source paper constructs a smooth renorming $\mathcal{X}$ of c_0 for which $\text{NA}(\mathcal{X}, \mathbb{R})$ contains no non-trivial cone. The paper then asks the following question on page 11.

Let us mention that it is still not known whether finite-rank operators can be always approximated by norm-attaining ones, even in the case when the range space is the two-dimensional Hilbert space ℓ_2^2 (in the words of Johnson and Wolfe [13, Question 6], the latter is *the most irritating open problem about norm-attaining operators*). Actually, it is still unknown if there is a Banach space X such that $\text{NA}(X, \ell_2^2)$ does not contain rank-two operators. Here is where the space $\mathfrak{X}$ comes into play in this setting. It is known that a finite rank operator $T \in \mathcal{L}(X, Y)$ belongs to $\text{NA}(X, Y)$ whenever $[\ker T]^\perp \subseteq \text{NA}(X, \mathbb{R})$ [17, Corollary 13.6]. Hence there are elements in $\text{NA}(X, \ell_2^2)$ of rank two whenever $\text{NA}(X, \mathbb{R})$ contains two-dimensional subspaces. Moreover, it is enough that $\text{NA}(X, \mathbb{R})$ contains non-trivial cones to get that $\text{NA}(X, Y)$ contains rank-two operators for every Banach space Y of dimension greater than or equal to two [17, Corollary 13.10], and this applies to all previously constructed Read norms, as explained in the introduction, see [17, Example 13.13]. However, $\text{NA}(\mathfrak{X}, \mathbb{R})$ does not contain non-trivial cones, hence no known result implies that $\text{NA}(\mathfrak{X}, \ell_2^2)$ contains rank-two operators.

Open Problem 7. *Does $\text{NA}(\mathfrak{X}, \ell_2^2)$ contain rank-two operators?*

Let us mention that [17, Theorem 13.24] contains a very general sufficient condition on $\text{NA}(X, \mathbb{R})$ (which implies that it contains subspaces of arbitrary large dimension) to assure that every finite-rank operator whose domain is X can be approximated by norm-attaining ones.

In words, Open Problem 7 asks whether

$$\text{NA}(\mathcal{X}, \ell_2^2)$$

contains a rank-two operator.

Candidate Partial Result

The result below does not solve Open Problem 7. It proves that, for each fixed two-codimensional kernel, norm-attainment is a nowhere dense condition in the corresponding finite-dimensional operator fibre.

Theorem 1. *Let $\mathcal{X}$ be Martin's space from arXiv:2406.07273, and let $H = \ell_2^2$ be the real two-dimensional Hilbert space. Let M be a closed subspace of $\mathcal{X}$ with $\text{codim } M = 2$, and set*

$$\mathcal{L}_M = \{T \in \mathcal{L}(\mathcal{X}, H) : M \subseteq \ker T\}.$$

Then

$$\text{NA}(\mathcal{X}, H) \cap \mathcal{L}_M$$

is nowhere dense in $\mathcal{L}_M$.

Corollary 1. *If $T \in \text{NA}(\mathcal{X}, \ell_2^2)$ has rank two and kernel M , then for every $\varepsilon > 0$ there is a rank-two operator $S : \mathcal{X} \rightarrow \ell_2^2$ such that*

$$\ker S = M, \quad \|S - T\| < \varepsilon, \quad S \notin \text{NA}(\mathcal{X}, \ell_2^2).$$

Definitions and Source Facts

Let $q : \mathcal{X} \rightarrow E := \mathcal{X}/M$ be the quotient map. Since $\text{codim } M = 2$, the space E is two-dimensional. The fibre $\mathcal{L}_M$ is naturally isometric to $\mathcal{L}(E, H)$: every $T \in \mathcal{L}_M$ has the form $T = Aq$ for a unique $A \in \mathcal{L}(E, H)$.

We use two facts from Martin's paper.

First, Proposition 2.4 in the source paper implies that

$$C_M := q(B_{\mathcal{X}}) \cap S_E$$

contains at most two antipodal pairs. In particular, C_M is finite.

Second, $\mathcal{X}$ is smooth. Equivalently, $\mathcal{X}^*$ is strictly convex. Since

$$E^* = (\mathcal{X}/M)^* \cong M^\perp$$

is a subspace of the strictly convex space $\mathcal{X}^*$, the dual E^* is strictly convex. Therefore the quotient E is smooth.

The Finite-Dimensional Lemma

Lemma 1. *Let E be a smooth real two-dimensional Banach space, let $H = \ell_2^2$, and fix $e \in S_E$. Define*

$$F_e = \{A \in \mathcal{L}(E, H) : \|Ae\| = \|A\|\}.$$

Then F_e is closed and nowhere dense in $\mathcal{L}(E, H)$.

Proof. Closedness is immediate from continuity of $A \mapsto \|A\|$ and $A \mapsto \|Ae\|$. We prove that F_e has empty interior.

Let $A \in F_e$ and let $\delta > 0$. If $A = 0$, choose any non-zero operator $B : E \rightarrow H$ with $Be = 0$ and $\|B\| < \delta$. Then $B \notin F_e$, so we are done.

Assume now that $A \neq 0$. Put $y = Ae$, so $\|y\| = \|A\| > 0$. Since E is smooth, there is a unique $j \in S_{E^*}$ such that $j(e) = 1$. Choose $d \in \ker j \setminus \{0\}$. Then

$$\left. \frac{d}{dt} \right|_{t=0} \|e + td\| = j(d) = 0.$$

Let $u = y/\|y\| \in S_H$. Since A attains its norm at e , we have

$$\|A(e + td)\| \leq \|y\| \|e + td\| \quad (t \in \mathbb{R}).$$

Differentiating this inequality at $t = 0$, from the right and from the left, gives

$$\langle u, Ad \rangle = 0.$$

Choose $B \in \mathcal{L}(E, H)$ with $Be = 0$ and $Bd = y$. For $\alpha > 0$, set

$$A_\alpha = A + \alpha B.$$

Then $A_\alpha e = y$, while

$$A_\alpha(e + td) = y + tAd + \alpha ty.$$

As H is Hilbert, the derivative at $t = 0$ of $\|A_\alpha(e + td)\|$ is

$$\langle u, Ad + \alpha y \rangle = \alpha \|y\| > 0.$$

The derivative at $t = 0$ of $\|e + td\|$ is 0, so for all sufficiently small positive t ,

$$\frac{\|A_\alpha(e + td)\|}{\|e + td\|} > \|y\| = \|A_\alpha e\|.$$

Hence e is not a norming point for A_α , and so $A_\alpha \notin F_e$. Taking $\alpha > 0$ small enough that $\|\alpha B\| < \delta$, we have found an operator outside F_e arbitrarily close to A . $\square$

Proof of Theorem 1

Identify $\mathcal{L}_M$ with $\mathcal{L}(E, H)$ by $T = Aq$. We claim that

$$\text{NA}(\mathcal{X}, H) \cap \mathcal{L}_M \subseteq \{0\} \cup \bigcup_{e \in C_M} F_e.$$

Indeed, let $T = Aq \in \text{NA}(\mathcal{X}, H) \cap \mathcal{L}_M$ be non-zero. Choose $x \in S_{\mathcal{X}}$ with

$$\|Aqx\| = \|T\| = \|A\|.$$

Put $e = qx$. If $\|e\| < 1$, then

$$\left\| A \frac{e}{\|e\|} \right\| = \frac{\|Ae\|}{\|e\|} > \|A\|,$$

which is impossible. Therefore $\|e\| = 1$, and since $x \in B_{\mathcal{X}}$, we have $e \in q(B_{\mathcal{X}}) \cap S_E = C_M$. Also

$$\|Ae\| = \|A\|,$$

so $A \in F_e$.

By Martin's Proposition 2.4, the set C_M is finite. By Lemma 1, each F_e is closed and nowhere dense in $\mathcal{L}(E, H)$. The singleton $\{0\}$ is also nowhere dense. Hence $\text{NA}(\mathcal{X}, H) \cap \mathcal{L}_M$ is contained in a finite union of closed nowhere dense subsets of $\mathcal{L}_M$, and is itself nowhere dense.

Verification Notes

The key perturbation lemma uses smoothness of E . This is why the special geometry of Martin's space is relevant: the constructed $\mathcal{X}$ is smooth, so $\mathcal{X}^*$ is strictly convex; strict convexity passes to the subspace $M^\perp \cong E^*$, and therefore E is smooth.

The conclusion is intentionally fixed-kernel. It does not prove that $\text{NA}(\mathcal{X}, \ell_2^2)$ has no rank-two operators, nor that such operators are not dense when kernels are allowed to vary. It only proves that every fixed two-dimensional quotient has a meagre set of norm-attaining Hilbert-plane operators.

Literature and Novelty Check

The proof uses the source paper's Proposition 2.4 and smoothness properties of $\mathcal{X}$. A targeted search on June 13, 2026 for phrases around $\text{NA}(X)$ containing non-trivial cones, Open Problem 7, BPBp/RSE tools, and rank-two operators into ℓ_2^2 found Martin's arXiv:2406.07273, KLMW's arXiv:1905.08272, and later BPBp/RSE papers, but no statement of the fixed-kernel nowhere-dense consequence above.

Human review should focus on Lemma 1 and on the passage from Martin's finite quotient-sphere set C_M to the operator fibre $\mathcal{L}_M$.